

AWS checklist and maintenance guide

Station name	SCO-V	
People present	MCIT TORA	instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency	
Date	10/AUG/2014 11:45	
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %: 10	

Notes:

Airport could not be kept open
later than 16:00 and there
was a charter scheduled first in the
morning.

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Metadata before maintenance (if establishing a new AWS: skip this page)

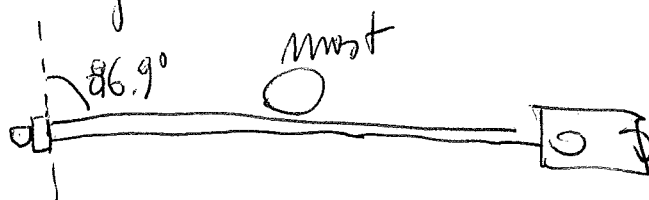
AWS leads

Time difference between logger clock and UTC	AWS + 2m16 +/-
Adjust logger time to UTC (be sure PC runs on UTC time)	☒ Y / ☐ N
Download data to PC and/or change CF Card (wait for green light)	☒ Y / ☐ N
Name of logger program	PROMICE 2010 B/N/17
Photo or screen dump of values in fast scan mode	☒ Y / ☐ N
Latitude (dd mm.mmm)	see GPS
Longitude (dd mm.mmm)	see GPS
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / ☐ N
Mast tilt in boom direction (+ if radiometer tilting down)	+87.2°
Mast tilt across boom looking from radiometer (- if clockwise)	-89.1°
Boom direction relative to north	320° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	☒ Y / ☐ N
Radiometer aligned with mast and inclinometer? If not, measure.	☒ Y / ☐ N
Temperature / humidity height (bottom of casing – surface)	240 cm
Sonic ranger on AWS height (membrane – surface)	238 cm
Wind sensor height (center of propeller – surface)	287 cm
Sonic ranger on stakes height (membrane – surface)	faller cm
Free length of stake – outer, holding boom with sonic ranger	484 cm
Free length of stake – middle	502 cm
Free length of stake – outer, holding boom without sonic ranger	581 cm
Free length of ablation hose on/above the ice surface	19 m 38 cm
Alt: depth of ablation hose using markings on hose	— cm
Height of ablation hose fluid level above the ice surface	155 cm
Vertical difference in surface height (at station – at hose) (approx)	+ 10 cm
Length of thermistor string on surface from surface marking	3 m 52 cm
Height of surface irregularities	50 cm
Stake assembly still standing?	☒ Y / ☒ N more or less
Apparent damage to sensors? If yes, which?	☒ Y / ☐ N

Notes:

radiometer was ~~loose~~ loose on fixing rod. if tightened, it was 85.9° CW when looking from the radiometer, compared to the tilt meter.

Wind monitor was not aligned.



Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	110722 →		- Aligned with mast: Y/N CORRECTED - Clean Y/N
Inclinometer	64 →		- Aligned with radiometer within 0.5°: Y/N NO, CORRECTED
Satellite antenna	650505 →		
Wind sensor	117286 →		- Box on sensor exactly towards mast (south): Y/N CORRECTED
Temperature / humidity assembly	62012 →		
T/hum assembly casing	60946546 →		- Fan spins and sounds OK: Y/N
T/hum assembly bracket	1111 →		(60812337)
Temperature probe in T/hum assembly	60946544 →		
HygroClip in T/hum assembly	60837623 →		- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS	2041 5939		- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel	C10903021071134 →		- Output OK: Y/N - Clean Y/N
GPS antenna	UPREAR.		- Stuck to top of logger enclosure: Y/N
Data logger	E1461 →		- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader	3493 →		- LED green of flashing orange (active): Y/N
Multiplexer	E4072 →		- Clicking as it should: Y/N
Iridium modem IMEI number	300034012832110 →		- Transmissions tested: Y/N - Problems: Y/N
Barometer	3472056 →		
Sonic ranger on stakes	3843 →	SAME SENSORS SAME MEM.	- Old sensor with new membrane: Y/N OLD W/ OLD MEMBER - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string	007 B25		- Old string left in ice: Y/N /cut
Ablation hose assembly	5005ABS →		- Old hose left in ice: Y/N /cut - Air in hose: Y/N removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 8 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

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Metadata after maintenance / establishment

Time logger clock set (to UTC)?	☒ Y / ☐ N
Name of logger program	2014d
Photo or screen dump of values in fast scan mode	☒ Y / ☐ N
Latitude (dd mm.mmm)	72.39378 N
Longitude (dd mm.mmm)	27.24441 W
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	+ 2.7 °
Mast tilt across boom looking from radiometer (- if clockwise)	- 1.2 °
Boom direction relative to north (radiometer should point to true south)	316 ° true (magn)
Box on wind sensor exactly towards mast? If not, change.	☒ Y / ☐ N
Radiometer aligned with mast and inclinometer? If not, align.	☒ Y / ☐ N
Temperature / humidity height (bottom of casing – surface)	232 232 cm
Sonic ranger on AWS height (membrane – surface)	232 232 cm
Wind sensor height (center of propeller – surface)	232 230 cm
Sonic ranger on stakes height (membrane – surface)	78 cm
Free length of stake – outer, holding boom with sonic ranger	116 cm
Free length of stake – middle	109 cm
Free length of stake – outer, holding boom without sonic ranger	124 cm
Free length of ablation hose on/above the ice surface	SAME cm
Alt: depth of ablation hose using markings on hose	— cm
Height of ablation hose fluid level above the ice surface	SAME cm
Vertical difference in surface height (at station – at hose) (approx)	232 SAME cm
Length of thermistor string on surface from surface marking	54 cm
Height of surface irregularities	SAME cm

Notes: